

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 03: TISSUES: PLANT & ANIMAL STRUCTURES

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: HISTOLOGICAL EXAMINATION OF PLANT STEMS (T.S. OF DICOT STEM)

Students prepared a thin transverse section (T.S.) of a young sunflower stem using a sharp razor blade. They floated the thin sections in water, transferred the thinnest uniform slice to a watch glass, stained it with Safranin and Fast Green, and mounted it in glycerin.

Under the microscope, they observed concentric tissue layers from outside to inside:

1. Outer epidermis covered with a cuticular layer.
2. Collenchymatous hypodermis at the corners.
3. Parenchymatous cortex with intercellular spaces.
4. A ring of vascular bundles with sclerenchymatous bundle caps (hard bast).
5. Vascular bundles showing xylem towards the center and phloem towards the periphery, separated by cambium.

Q.1 Which tissue in the hypodermis provides mechanical flexibility, allowing young stems to bend without breaking?

- | | |
|------------------|-----------------|
| (A) Parenchyma | (B) Collenchyma |
| (C) Sclerenchyma | (D) Aerenchyma |

Q.2 How can sclerenchyma fibers in the bundle cap be distinguished microscopically from parenchyma cells?

- | | |
|--|---|
| (A) Sclerenchyma cells are living with thin cellulose walls and large chloroplasts | (B) Sclerenchyma cells are dead at maturity with narrow lumens and heavily lignified, thickened secondary walls |
| (C) Sclerenchyma cells have large intercellular spaces filled with air | (D) Sclerenchyma cells contain prominent central nuclei |

Q.3 Which meristematic tissue strip present between xylem and phloem is responsible for secondary radial growth (increase in girth)?

- | | |
|---|--------------------------|
| (A) Apical meristem | (B) Intercalary meristem |
| (C) Vascular cambium (Lateral meristem) | (D) Protoderm |

Q.4 Which constituent of xylem tissue is the only living cell type present in mature dicot xylem?

- | | |
|----------------------|-------------------|
| (A) Xylem tracheids | (B) Xylem vessels |
| (C) Xylem parenchyma | (D) Xylem fibres |

Q.5 Why does xylem stain intense red with Safranin stain?

- | | |
|--|---------------------------------|
| (A) Safranin reacts selectively with lignin present in xylem tracheid and vessel walls | (B) Safranin stains only lipids |
| (C) Xylem absorbs chlorophyll | (D) Xylem is rich in starch |

CASE STUDY 2: STOMATAL DISTRIBUTION & TRANSPIRATION INVESTIGATION

A botany student examined the distribution of stomata on the dorsal (upper) and ventral (lower) surfaces of a dorsiventral dicot leaf (*Hibiscus*).

- She prepared epidermal peels from both leaf surfaces and counted stomata in five microscopic fields of view at 400x magnification.
- **Upper Epidermis:** Average 12 stomata per field.
- **Lower Epidermis:** Average 85 stomata per field.

To verify the physiological consequence of this unequal distribution, she attached blue dry Cobalt Chloride (CoCl_2) paper strips to both surfaces of an intact living leaf using glass slides and clips, placed the plant in sunlight, and timed the color change from blue to pink.

Q.6 On which leaf surface did the Cobalt Chloride paper turn pink first?

- | | |
|--|---|
| (A) The upper surface, because it receives direct sunlight | (B) The lower surface, because it contains a much higher density of stomata resulting in faster transpiration |
| (C) Both surfaces turned pink at the exact same instant | (D) Neither surface changed color |

Q.7 What is the evolutionary biological advantage of having fewer stomata on the upper surface of terrestrial dicot leaves?

- | | |
|---|---|
| (A) To prevent insects from eating the leaf | (B) To reduce excessive water loss caused by direct solar irradiation and wind exposure |
| (C) To block absorption of sunlight | (D) To reduce oxygen intake |

Q.8 Which specialized epidermal cells control the opening and closing of each stomatal pore?

- | | |
|---------------------------|-----------------|
| (A) Subsidiary parenchyma | (B) Guard cells |
| (C) Trichomes | (D) Lenticels |

Q.9 What physiological change causes a stomatal pore to open during the day?

- | | |
|--|---|
| (A) Guard cells lose water and become flaccid | (B) Guard cells absorb water via endosmosis, become turgid, and their thin outer walls stretch outwards, pulling open the thick inner walls |
| (C) Guard cells shrink and detach from the epidermis | (D) The pore fills with starch grains |

Q.10 What protective, hydrophobic chemical substance coats the outer surface of the leaf epidermis to minimize water loss?

- | | |
|-------------|---------------------------------|
| (A) Suberin | (B) Cutin (forming the cuticle) |
| (C) Pectin | (D) Chitin |

CASE STUDY 3: PHLOEM TRANSLOCATION VS XYLEM ASCENT OF SAP

In a classical plant physiology investigation, students performed two experiments:

- **Experiment A (Dye Tracing):** A leafy branch of a balsam plant was placed in a beaker containing water stained with red eosin dye for 3 hours. Thin cross-sections of the stem revealed that only vascular xylem vessels were stained red.
- **Experiment B (Girdling / Ringing):** A complete ring of bark (outer tissue including phloem down to the cambium) was removed from the woody stem of a potted plant. After several weeks, a distinct swelling developed immediately above the girdle, while tissues below remained unthickened.

Q.11 What conclusion is drawn from Experiment A regarding xylem tissue?

- | | |
|--|---|
| (A) Xylem conducts organic sugars downward from leaves | (B) Xylem is the vascular tissue responsible for unidirectional upward conduction of water and dissolved minerals (ascent of sap) |
| (C) Xylem stores lipids in the bark | (D) Xylem transports gases to roots |

Q.12 Why did a swelling develop immediately above the girdle in Experiment B?

- | | |
|---|--|
| (A) Water accumulated because xylem was blocked | (B) Downward transport of photosynthesized organic nutrients (sucrose) was halted by removing phloem, causing food accumulation and cell division above the girdle |
| (C) Bacterial infection attacked the exposed wood | (D) Turgor pressure caused the stem to stretch |

Q.13 Which constituent of phloem tissue lacks a cell nucleus at maturity yet remains living and metabolically functional?

- | | |
|-----------------------|-------------------------|
| (A) Phloem parenchyma | (B) Sieve tube elements |
| (C) Companion cells | (D) Phloem fibers |

Q.14 Which phloem component regulates the metabolic transport activities of an adjacent enucleated sieve tube element?

- | | |
|--------------------|------------------|
| (A) Companion cell | (B) Xylem vessel |
| (C) Tracheid | (D) Guard cell |

Q.15 Which driving physical force is primarily responsible for the rapid ascent of sap in tall trees during sunny daylight hours?

- | | |
|------------------------------------|------------------------------------|
| (A) Transpirational pull | (B) Root pressure alone |
| (C) Atmospheric pressure on leaves | (D) Active pumping by xylem fibers |

CASE STUDY 4: COMPARATIVE HISTOLOGY OF ANIMAL MUSCULAR TISSUES

Students observed three prepared stained histological slides of mammalian muscle tissues under high power (400x) magnification:

- **Slide 1:** Long, unbranched, cylindrical fibers with alternating dark and light cross-striations and multiple flattened nuclei located peripherally beneath the sarcolemma.
- **Slide 2:** Spindle-shaped (fusiform) unbranched fibers with pointed ends, lacking striations, each having a single centrally positioned oval nucleus.
- **Slide 3:** Cylindrical, branched fibers showing light striations, single central nuclei, and dark transverse junctional bands called intercalated discs.

Q.16 Identify the muscular tissue observed on Slide 1:

- | | |
|----------------------------|--------------------------------|
| (A) Smooth visceral muscle | (B) Skeletal (striated) muscle |
| (C) Cardiac muscle | (D) Adipose tissue |

Q.17 Where in the human body is the muscular tissue of Slide 2 (spindle-shaped, unstriated) typically located?

- | | |
|---------------------------------|---|
| (A) Biceps and quadriceps limbs | (B) Walls of internal hollow organs such as stomach, intestine, and blood vessels |
| (C) Heart wall (myocardium) | (D) Tongue and pharynx |

Q.18 What is the critical physiological role of the intercalated discs observed on Slide 3 (Cardiac muscle)?

- | | |
|---|--|
| (A) They store glycogen granules | (B) They contain gap junctions that allow electrical impulses to spread rapidly, enabling synchronized rhythmic contractions of the heart chambers |
| (C) They anchor muscles to skeletal bones | (D) They secrete adrenaline hormone |

Q.19 Why does skeletal muscle experience fatigue after prolonged vigorous exercise, whereas cardiac muscle contracts tirelessly throughout life?

- | | |
|--|---|
| (A) Skeletal muscle has more blood vessels than cardiac muscle | (B) Cardiac muscle has an extraordinarily rich capillary supply, high mitochondrial density, and relies strictly on aerobic metabolism without accumulating lactic acid |
| (C) Skeletal muscle lacks ATP | (D) Cardiac muscle contracts only when a person is awake |

Q.20 Which of the three muscle types is categorized as under voluntary conscious nervous control?

- | | |
|--------------------------|------------------------|
| (A) Skeletal muscle only | (B) Smooth muscle only |
| (C) Cardiac muscle only | (D) All three types |

CASE STUDY 5: CONNECTIVE TISSUE MECHANICS & HUMAN BLOOD CYTOLOGY

During a sports medicine workshop, a physiotherapist discussed connective tissue injuries and microscopic hematology:

- **Patient X:** A footballer suffered a severe joint sprain; MRI revealed a torn ligament at the knee.
- **Patient Y:** A runner experienced acute heel pain diagnosed as Achilles tendinitis.
- **Blood Smear Analysis:** A normal stained blood film examined under oil immersion magnification displayed biconcave enucleated red cells, white cells (neutrophils, lymphocytes, monocytes), and tiny cell fragments.

Q.21 How do ligaments fundamentally differ from tendons in terms of their anatomical connections and fiber composition?

- | | |
|--|--|
| (A) Ligaments connect bone to bone and contain elastic yellow fibers; tendons connect muscle to bone and are composed of inelastic white collagen fibers | (B) Ligaments connect muscle to bone; tendons connect bone to bone |
| (C) Ligaments are dead; tendons are living | (D) Ligaments contain calcium phosphate; tendons contain liquid plasma |

Q.22 Why do ligament tears heal much more slowly than fractured bones?

- | | |
|---------------------------------------|--|
| (A) Ligaments contain no living cells | (B) Bone has a rich vascular blood supply through Haversian canals, whereas dense fibrous ligament tissue has minimal blood supply |
| (C) Bones are softer than ligaments | (D) Ligaments dissolve in body fluids |

Q.23 Which formed cellular element in blood is enucleated at maturity and specializes in oxygen transport via hemoglobin?

- | | |
|----------------|----------------------------------|
| (A) Neutrophil | (B) Erythrocyte (Red Blood Cell) |
| (C) Lymphocyte | (D) Platelet |

Q.24 Which blood component consists of cytoplasmic cell fragments derived from megakaryocytes that are essential for blood clotting?

- (A) Platelets (Thrombocytes) (B) Eosinophils
(C) Basophils (D) Monocytes

Q.25 Which connective tissue possesses a hard, rigid matrix composed of calcium and phosphorus compounds?

- (A) Hyaline cartilage
(B) Areolar tissue
(C) Bone tissue
(D) Adipose tissue

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	Collenchyma cells have localized pectin and cellulose thickenings at their cell corners, providing mechanical support and flexibility.
Q.2	(B)	Sclerenchyma consists of dead, elongated fiber cells with narrow obliterated lumens and thick walls impregnated with lignin.
Q.3	(C)	Vascular cambium is a lateral meristem whose periclinal divisions produce secondary xylem and phloem, increasing stem girth.
Q.4	(C)	Xylem parenchyma is the only living element; tracheids, vessels, and fibers are dead, hollow, lignified cells at functional maturity.
Q.5	(A)	Safranin is a basic histological dye that binds specifically to poly-phenolic lignin polymers present in xylem cell walls.
Q.6	(B)	The lower epidermis has over 7 times higher stomatal density. Greater water vapor transpires through the lower surface, rapidly turning anhydrous blue CoCl_2 paper into pink hydrated form.
Q.7	(B)	Positioning most stomata on the shaded lower surface reduces excessive transpirational water loss in hot, dry terrestrial environments.
Q.8	(B)	Each stomatal pore is flanked by two kidney-shaped (or dumb-bell shaped in grasses) guard cells that regulate pore diameter.
Q.9	(B)	When guard cells absorb water and swell, their differentially thickened walls (thin elastic outer wall, thick rigid inner wall) bow outward, opening the pore.
Q.10	(B)	Cutin is a waxy, hydrophobic lipid polymer secreted by epidermal cells to form the protective outer cuticle layer.
Q.11	(B)	The selective red staining of tracheids and vessels demonstrates that water and dissolved mineral solutes ascend through xylem.
Q.12	(B)	Girdling removes the phloem sieve tubes. Sugars synthesized by leaves above the ring cannot travel downward to roots, accumulating and causing cellular swelling.
Q.13	(B)	Mature sieve tube elements lack a nucleus, vacuole, and ribosomes, but remain alive through intimate plasmodesmatal connections with nucleated companion cells.
Q.14	(A)	Companion cells are specialized parenchymatous cells with dense cytoplasm and large nuclei that load and unload sucrose into sieve tubes.
Q.15	(A)	Transpiration of water vapor from leaf stomata generates a continuous negative hydrostatic pressure gradient (tension/pull) that draws continuous water columns up xylem.
Q.16	(B)	Skeletal muscle fibers are voluntary, cylindrical, multinucleate (syncytial), with peripheral nuclei and prominent cross-striations.
Q.17	(B)	Smooth (unstriated, involuntary) muscle fibers line visceral organs such as the gastrointestinal tract, urinary bladder, uterus, and blood vessels.
Q.18	(B)	Intercalated discs are specialized cell junctions containing desmosomes and gap junctions that electrically couple adjacent cardiomyocytes for coordinated contraction.
Q.19	(B)	Cardiomyocytes have abundant mitochondria (up to 40% of cell volume) and continuous oxygen supply, preventing anaerobic fatigue.
Q.20	(A)	Only skeletal muscle contractions can be initiated voluntarily via the somatic nervous system; smooth and cardiac muscles are involuntary.
Q.21	(A)	Ligaments join bone to bone with high elasticity (elastin fibers); tendons join muscle to bone with high tensile strength and limited elasticity (dense collagen).

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.22	(B)	Bone is richly vascularized with osteocytes continuously remodeled by blood flow; dense fibrous connective tissues like ligaments are poorly vascularized.
Q.23	(B)	Mammalian erythrocytes lose their nucleus, mitochondria, and ER during maturation to maximize internal volume for oxygen-carrying hemoglobin molecules.
Q.24	(A)	Platelets are enucleated discoid fragments that aggregate at injury sites and release clotting factors to form hemostatic fibrin plugs.
Q.25	(C)	Bone matrix is composed of collagen fibers impregnated with mineral hydroxyapatite salts ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$), conferring compressive strength.